11.4

DATA SET DS1123

24. Where to Shop? Do you shop at the grocery store closest to home or do you look for the store that has the best prices? We compared the regular prices at four different grocery stores for eight items purchased on the same day.

Items	Stores			
	Vons	Ralphs	Stater Bros	WinCo
Salad mix, 360 g bag	3.99	2.79	1.99	1.78
Hillshire Farm® Beef Smoked Sausage, 420 g	4.29	4.29	3.99	2.50
Kellogg's Raisin Bran®, 750 g	4.49	5.49	4.49	3.15
Kraft® Philadelphia® Cream Cheese, 250 g	2.99	3.19	2.79	1.48
Kraft® Ranch Dressing, 450 mL	3.19	3.49	3.49	1.48
Treetop® Apple Juice, 2 L	2.99	3.49	3.49	1.58
Dial® Bar Soap, Gold, 125 g	5.99	6.49	5.79	5.14
Jif® Peanut Butter, Creamy, 850 g	5.15	5.49	4.79	4.34

a. What are the blocks and treatments in this experiment?

b. Do the data provide evidence to indicate that there are significant differences in prices from store to store? Support your answer statistically using the ANOVA printout that follows.

c. Are there significant differences from block to block? Was blocking effective?

ANOVA: Price versus Item, Store

Analysis of Variance for Price

Source	DF	SS	MS	F	P
Store	3	13.192	4.3974	26.52	0.000
Item	7	40.995	5.8564	35.32	0.000
Error	21	3.482	0.1658		
Total	31	57.669			

Model Summary

S	R-sq	R-sq(adj)
0.407181	93.96%	91.09%

a. Blocks: Items
treatments: stores.

b. Yes. $F = 26.52$, $p = 0$
$p < 0.05$, rej H_0 #

c. Yes. $F = 35.32$, $p = 0$
$p < 0.05$, effective.

25. Where to Shop?, continued Refer to Exercise 24. The printout that follows provides the average costs of the selected items for the $k = 4$ stores.

Means

	Store	N	Price
1	Ralphs	8	4.34000
2	Stater Bros	8	3.85250
3	Vons	8	4.13500
4	WinCo	8	2.68125

a. What is the appropriate value of $q_{.05}(k, df)$ for testing for differences among stores?

b. What is the value of $\omega = q_{.05}(k, df)\sqrt{\frac{MSE}{b}}$?

c. Use Tukey's pairwise comparison test among stores used to determine which stores differ significantly in average prices of the selected items.

a. $q_{0.05}(4,21) \approx 3.96$

b. $\omega = 3.96\sqrt{\frac{0.1658}{8}}$

$= 0.57$

c.

1 − 2	0.48875	No
1 − 3	0.2050	No
1 − 4	1.6588	Yes
2 − 3	0.2825	No
2 − 4	1.1719	Yes
3 − 4	1.4538	Yes

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26. Rocket Propellants An experiment was conducted to compare three mixtures of components in making rocket propellant. Five samples of each mixture were obtained for testing. Five investigators measured the propellant thrust in pounds for each of the mixtures, with the mixtures presented in random order.

DATA SET DS1124

	Mixtures		
Investigators	1	2	3
1	42	39	45
2	37	36	40
3	43	48	50
4	41	47	52
5	39	42	47

a. Identify the treatments and the blocks in this experiment and provide an appropriate analysis of variance.

b. Is there sufficient evidence to conclude there is a significant difference in the propellant thrust for the mixtures? Use $\alpha = .01$.

c. Was blocking effective? Use the p-value to draw your conclusions and justify your answer.

d. Use Tukey's pairwise comparisons with $\alpha = .01$ to determine which propellant mixtures differed from others. Summarize your results.

a. treatments = 3種 propellant mixtures
blocks = 5位 investigators

ANOVA	df	SS	MS	F
Mixtures	2	107.2	53.6	0.005
Investigators	4	176.4	44.1	0.005
Error	8	38.8	4.85	
total	14	322.4		

b.
Yes, $F = 11.05$, $p = 0.005 < 0.01$
reject Ho #

c. Yes, Blocking was effective

$F = 9.09$, $p = 0.005 < 0.01$

d. Tukey's $\alpha = 0.01$

① = 40.14 $\quad HSD = 5.155$

② = 42.14

③ = 46.8

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